

Jose Caraveo

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EDUCATION

University of Colorado Boulder

Bachelor of Arts in Computer Science, Mathematics (Dual Degree)

May 2028 (Expected)

RELEVANT COURSEWORK

Data Structures, Computer Systems, Linear Algebra, Discrete Math, Multivariable Calculus, Probability Theory, Numerical Analysis, Database Design & Analysis, Mathematical Statistics

EXPERIENCE

Software Engineer Co-op Intern

September 2026 -

RTX Corporation

- Joining the Intelligence & Space team

Research Intern

May 2025 – August 2025

Bioastronautics Lab, University of Colorado Boulder

- Integrated machine learning components using Python, Scikit-learn, and PyTorch for preprocessing, feature extraction, model training, and evaluation
- Consolidated multiple ML/DL classification pipelines into a unified Python repository, improving experiment reproducibility and workflow organization
- Sourced and evaluated external datasets and lab data to support null boosting class for model training
- Analyzed academic literature on G-LOC prediction to guide model parameter selection and experimental design
- Developed a config-driven pipeline controller (config.json) for configuring classifiers, sample sizes, imbalance techniques, and hyperparameter optimization trials

Undergraduate Research Assistant

Dec. 2024 – May 2025

Human Interaction and Robotics Group, University of Colorado Boulder

- Co-authored a research paper published to the IEEE RO-MAN conference
- Reviewed academic literature to support ongoing robotics research and project planning

PROJECTS

Sampling Profiler | C++, POSIX signals

- Built a CPU sampling profiler in C++ that captures call stacks at 100 Hz using POSIX interval timers and SIGPROF, unwinding the stack from saved register context within an async-signal-safe handler
- Symbolized captured addresses against the binary's symbol table and aggregated samples into folded-stack format, producing flamegraphs that correctly attributed 99% of runtime to the expected hot path

Escape the Submarine | Lua, Roblox API

- Reached 2,000+ plays with a Roblox game built in Lua, implementing teleportation, collectible tracking backed by a persistent datastore, and player-facing UI

TECHNICAL SKILLS

Languages: C++, Python, Java, SQL

Libraries/Frameworks: PyTorch, Scikit-learn, pandas, NumPy, Optuna, Matplotlib, MongoDB

Systems / Tools: Linux, Git, PyCharm

Interests: Weightlifting, History, Combat sports, basketball

PUBLICATIONS

Employing Laban Shape for Generating Emotionally and Functionally Expressive Trajectories in Robotic Manipulators. RO-MAN 2025 [Link](#)

STUDENT SOCIETIES & CERTIFICATIONS

GFACT Cybersecurity Certificate (GIAC, 2023)

Member, Service and Social Action LLC (2025-26)